Project - Low Level Design

On

Multi-Agent Hospitality System

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Table of Contents

1. Introduction
    a. Scope of the document
    b. Intended Audience
    c. System overview
2. System Design
    a. Application Design
    b. Tool Design
    c. Process Flow
    d. Information Flow
    e. Components Design
    f. Key Design Considerations
    g. API Catalogue
3. References

1. Introduction

1.1 Scope of the Document

This document outlines the low-level design of the **Multi-Agent Hospitality System**. It provides a detailed description of the system's internal architecture, components, data flow, and interactions between different modules. The document covers the design of the user interface, backend processing using a multi-agent framework, integration with the Groq API for generating intelligent responses, and supporting utilities such as logging and external data fetching. It also highlights the structure of the application, including process flow, information flow, and API usage, ensuring a clear understanding of how the system operates at a detailed level. This document serves as a technical reference for developers and stakeholders involved in the development, evaluation, and maintenance of the system.

1.2 Intended Audience

This document is intended for developers, system designers, and evaluators involved in the Multi-Agent Hospitality System. It helps in understanding the system's structure, design, and functionality, and serves as a reference for implementation, evaluation, and future enhancements.

1.3 System overview

The Multi-Agent Hospitality System is a web-based application that generates personalized travel itineraries based on user input such as destination, number of days, and budget. It uses multiple AI agents to perform tasks like research, planning, budgeting, and formatting, and provides a structured day-wise travel plan to the user.

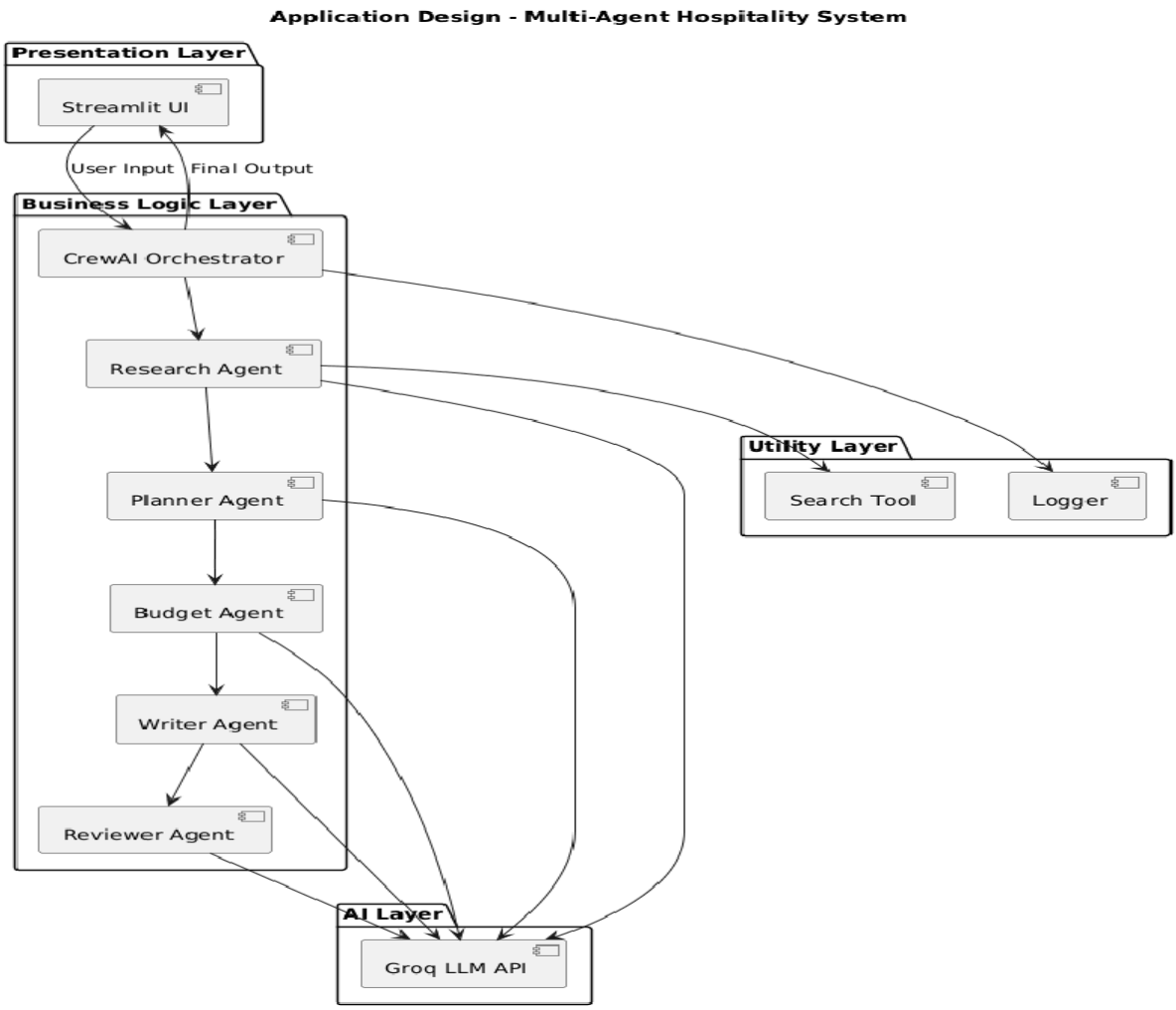
2. System Design:

2.1 Application Design

- The Multi-Agent Hospitality System follows a layered architecture to ensure modularity, scalability, and ease of maintenance. The application is divided into

four main layers: the Presentation Layer, Business Logic Layer, AI Layer, and Utility Layer.

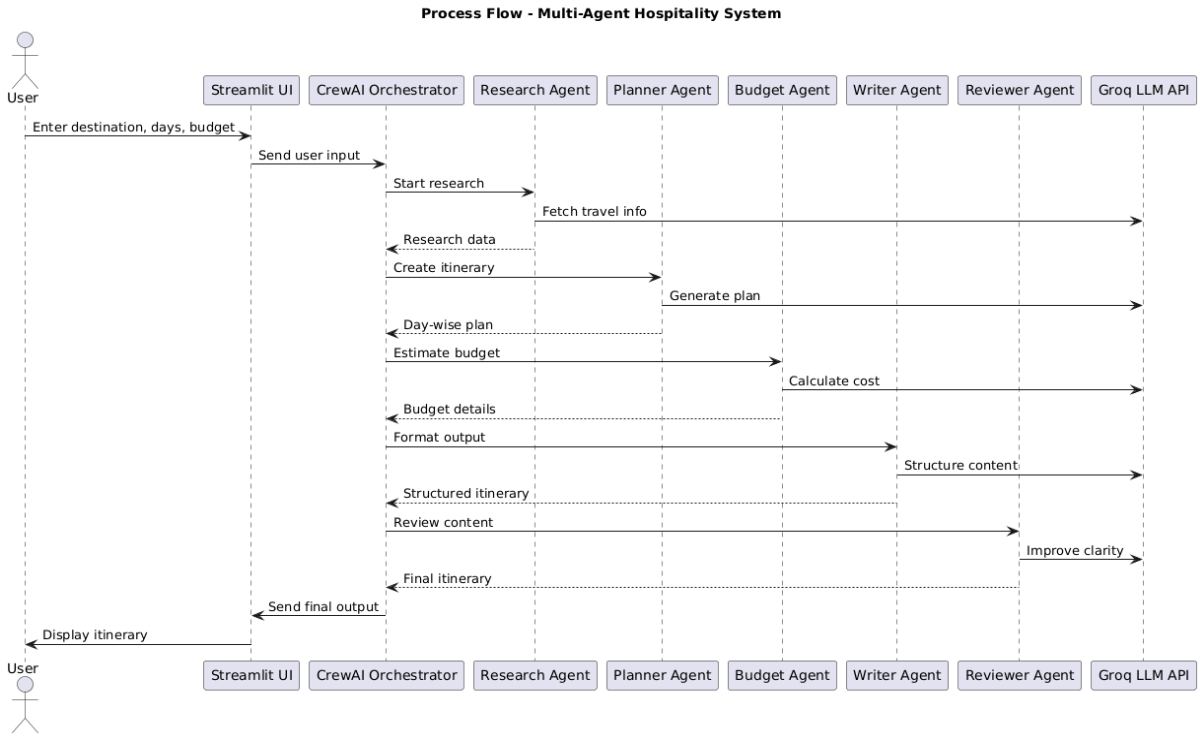
- The Presentation Layer is implemented using Streamlit and is responsible for handling user input and displaying the generated travel itinerary. The Business Logic Layer uses CrewAI to manage multiple specialized agents, each performing a specific task such as research, planning, budgeting, writing, and reviewing.
- The AI Layer integrates the Groq LLM API, which is used by all agents to generate intelligent and context-aware responses. The Utility Layer includes supporting components such as logging for tracking system activity and search tools for fetching additional travel-related information.
- This layered approach ensures clear separation of concerns, making the system more organized, flexible, and easier to extend in the future.



2.2 Process Flow:

The process flow of the Multi-Agent Hospitality System describes how user input is processed step-by-step to generate a travel itinerary.

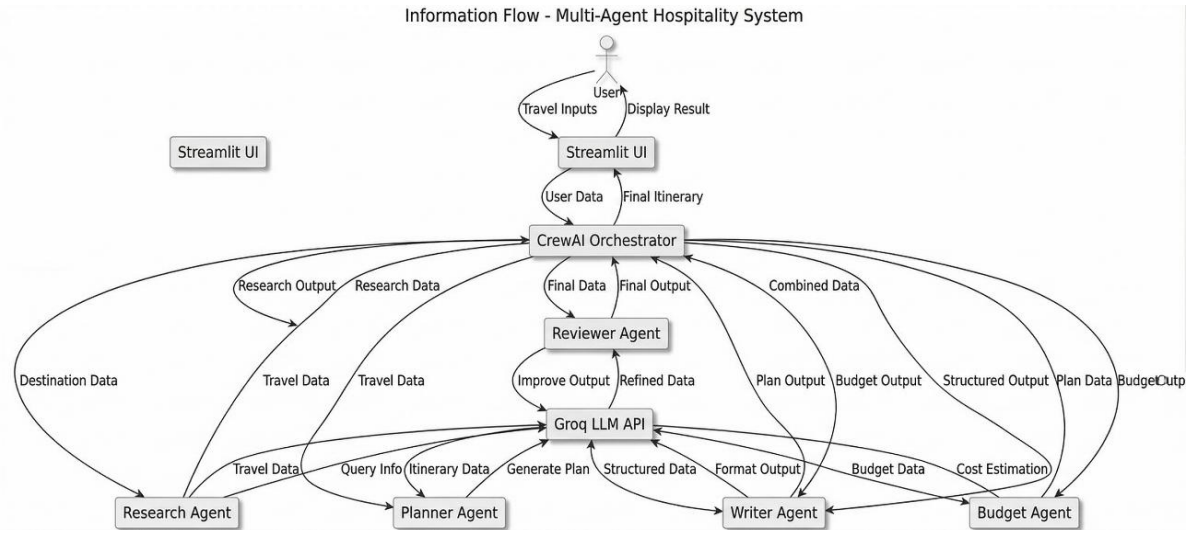
User → Streamlit UI → CrewAI Orchestrator → Research Agent → Groq LLM → Planner Agent → Groq LLM → Budget Agent → Groq LLM → Writer Agent → Groq LLM → Reviewer Agent → Groq LLM → CrewAI Orchestrator → Streamlit UI → User



2.3 Information Flow

The information flow in the Multi-Agent Hospitality System describes how data moves between different components during execution. The user provides input such as destination, number of days, and budget through the Streamlit UI. This input is forwarded to the CrewAI orchestrator, which distributes the information to various agents.

Each agent processes the data and generates intermediate outputs using the Groq LLM. These outputs are passed sequentially from one agent to another until the final itinerary is created. The processed information is then returned to the UI, where it is displayed to the user.



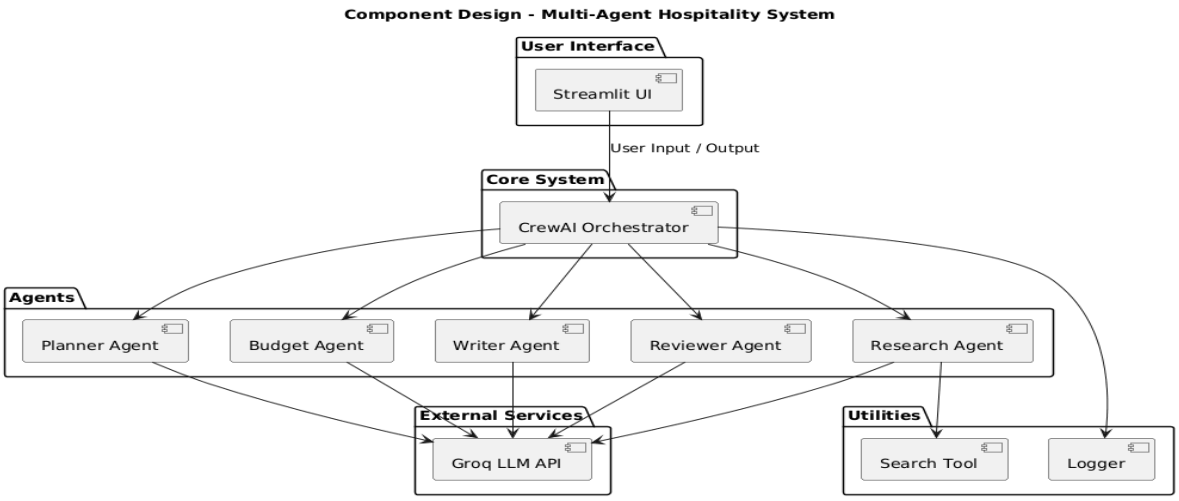
2.4 Components Design

The system follows a modular component-based architecture.

- Each component is responsible for a specific functionality.
- This ensures high cohesion and low coupling.
- The design makes the system easy to maintain and extend.
- Streamlit UI
    - Handles user input
    - Displays the generated itinerary
- CrewAI Orchestrator
    - Controls and manages the execution of all agents
- Agents (Core Components)
    - Research Agent → Collects travel information
    - Planner Agent → Creates day-wise itinerary
    - Budget Agent → Estimates travel cost
    - Writer Agent → Formats the output
    - Reviewer Agent → Improves final output
- Groq LLM API
    - Acts as the core intelligence provider
    - Used by all agents to generate responses
- Utility Components

- Logger → Stores system logs
- Search Tool → Fetches external travel data

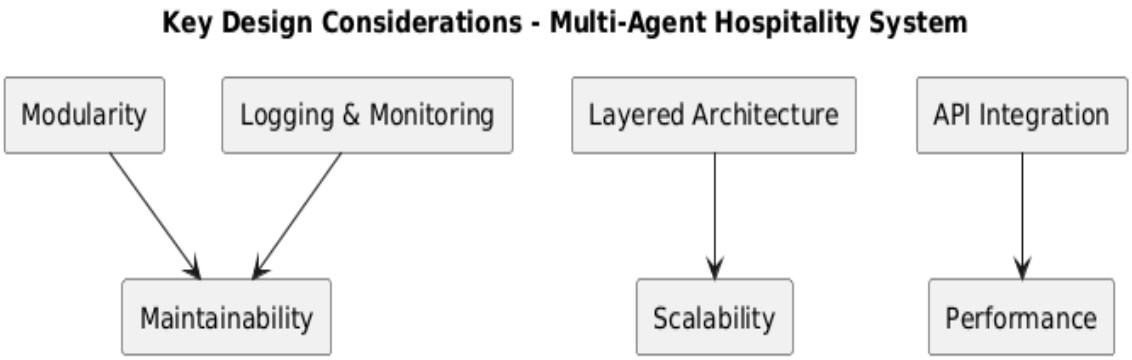
- All components work together in a structured flow to generate the final travel itinerary.



2.5 Key Design Considerations-

- The system is designed to ensure efficiency, scalability, and maintainability.
- A modular architecture is used where each agent performs a specific task such as research, planning, budgeting, writing, and reviewing.
- Separation of concerns is maintained, making the system easier to understand, maintain, and extend.
- A layered architecture is followed, dividing the system into presentation, business logic, AI, and utility layers.
- This layered approach reduces dependency between components and improves organization.
- The system is scalable, allowing new agents or features to be added without affecting existing functionality.
- Efficient integration with external APIs like the Groq LLM ensures smooth and reliable response generation.
- Logging is implemented to track system activities and errors for debugging and monitoring purposes.

- Performance is optimized by using lightweight models and sequential execution of agents.
- The design focuses on flexibility, maintainability, and efficient AI integration.



2.6 API Catalogue:

2.6.1 Groq LLM API

- Used to generate intelligent and context-aware travel responses.
- Handles tasks such as itinerary generation, cost estimation, and content formatting.
- Requires an API key for authentication.
- Used by all agents in the system.

2.6.2 DuckDuckGo Search API (Optional)

- Used to fetch travel-related information such as tourist places and food suggestions.
- Helps the Research Agent gather real-world data.
- Does not require strict authentication.

2.6.3 Internal CrewAI API

- Used to manage communication between agents.
- Handles task execution and data passing between different agents.
- Ensures sequential flow of operations.

3. References:

- Official documentation of **CrewAI** – for understanding multi-agent orchestration, agent roles, and workflow design.
- Official **Streamlit** documentation – for building interactive user interfaces and deploying data-driven applications.
- **Groq API** documentation – for integrating high-performance AI inference and model execution.
- Official **Python** documentation – for core programming concepts, libraries, and best practices.
- Curated online tutorials, research articles, and learning resources on **multi-agent systems** – for conceptual understanding and implementation strategies.